The documentation and process conversion measures necessary to comply with this document shall be completed by 6 July 2009.

INCH-POUND

MIL-PRF-19500/177G
6 April 2009
SUPERSEDING
MIL-PRF-19500/177F
4 August 1999

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON LOW-POWER
TYPES 2N1131, 2N1131L, 2N1132, 2N1132L, JAN, AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP silicon low-power transistors. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, 2N1131 and 2N1132 (TO-39), 2N1131L, and 2N1132L (TO-5).

* 1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

P_T (1) T_C = +25°C	P_T (2) T_A = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{OP} and T_J
W	W	V dc	V dc	V dc	mA dc	°C
2.0	0.6	50	40	5.0	600	-65 to +200

(1) Derate linearly 11.4 mW/°C for $T_C \geq$ +25°C.
(2) Derate linearly 3.4 mW/°C for $T_A \geq$ +25°C.

* 1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

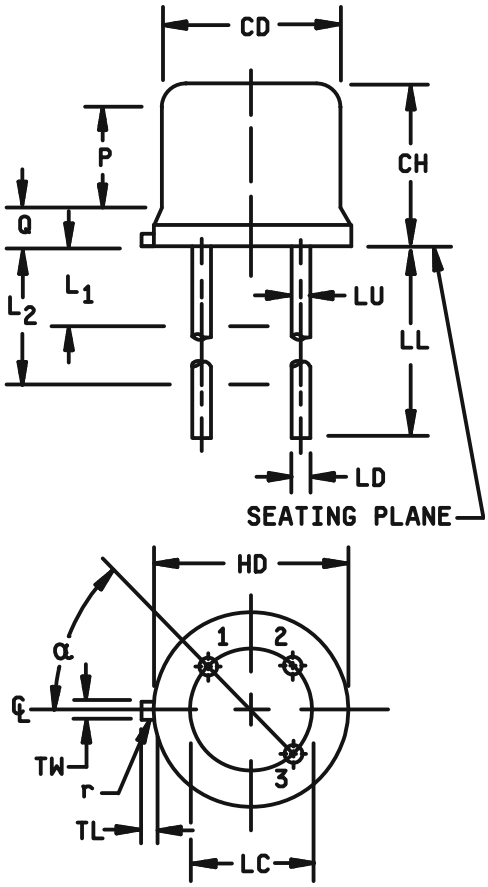
Types	h_{FE1} (1) V_{CE} = 10 V dc, I_C = 150 mA dc		h_{fe1} V_{CE} = 5.0 V dc, I_C = 1.0 mA dc, f = 1 kHz		$V_{BE(SAT)}$ (1) I_C = 150 mA dc, I_B = 15 mA dc		$V_{CE(SAT)}$ (1) I_C = 150 mA dc, I_B = 15 mA dc		C_{obo} f = 1 MHz, V_{CB} = 10 V dc, I_E = 0		$\lvert h_{fe} \rvert$ f = 20 MHz, I_C = 50 mA dc, V_{CE} = 10 V dc	
					V dc		V dc		pF			
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
2N1131, L	20	45	15	50		1.5		1.3		45	2.5	20
2N1132, L	30	90	30	90		1.5		1.3		45	3.0	20

(1) Pulsed (see 4.4.1).

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	6
CH	.240	.260	6.12	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		7
LD	.016	.019	0.41	0.48	8,9
LL	See note 14				
LU	.016	.019	0.41	0.48	8,9
L_1		.050		1.27	8,9
L_2	.250		6.35		8,9
P	.100		2.54		7
Q		.030		0.76	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		7
1, 2, 10, 12, 13, 14					



NOTES:

1. Dimensions are in inches.
* 2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
* 6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauging procedure.
8. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in and beyond LL minimum.
9. All three leads.
10. The collector shall be internally connected to the case.
11. Dimension r (radius) applies to both inside corners of tab.
* 12. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.
13. Lead 1 = emitter, lead 2 = base, lead 3 = collector.
14. For L-suffix or non-S-suffix devices (TO-5), dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max

* FIGURE 1. Physical dimensions 2N1131 and 2N1132 (TO-39), 2N1131L and 2N1132L (TO-5).

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

R_{BE}.....External resistance base to emitter.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

* 3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

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3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX level only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurements JANTX level only
(1) 3c	Thermal impedance method 3131 of MIL-STD-750 (see 4.3.2)
9	Not applicable
11	I_{CBO1} and h_{FE2}
12	See 4.3.1
13	ΔI_{CBO1} = 100 percent of initial value or 1 μA dc whichever is greater; Δh_{FE2} = ±20 percent of initial value; subgroup 2 of table I herein.

* (1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

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* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX quality level. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. Use method 3100 of MIL-STD-750 to measure T_J.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, and t_{SW} (V_C and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIb (JAN and JANTX) of MIL-PRF-19500 and as follows.. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
B3	1027	T_A = room ambient as defined in the general requirements of MIL-STD-750, V_{CB} = 30 V dc; adjust P_T to achieve T_J = 150°C min.
B3	2037	Test condition A.
B5	3161	For qualification only.
B6	1032	$T_{STG(max)}$ = +175°C

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E
C6	1026	T_A = room ambient as defined in the general requirements of MIL-STD-750, V_{CB} = 30 V dc; adjust P_T to achieve T_J = 150°C min.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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***** TABLE I. Group A inspection.

* Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal Impedance	3131	See 4.3.2.	$Z_{\theta JX}$			°C/W
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc
Collector - base breakdown voltage	3001	Bias condition D; I_C = 10 μA dc	$V_{(BR)CBO}$	50		V dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO1}		10	μA dc
Collector - emitter cutoff current	3041	Bias condition B; V_{CE} = 50 V dc; pulsed (see 4.5.1); $R_{BE} \leq 10$ ohms	I_{CER}		10	mA dc
Collector – base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO2}		1.0	μA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 5.0 V dc	I_{EBO}		100	μA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}			
2N1131, L				20	45	
2N1132, L				30	90	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 5.0 mA dc; pulsed (see 4.5.1)	h_{FE2}			
2N1131, L				15		
2N1132, L				25		
Collector - emitter saturated voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)}$		1.3	V dc
Base – emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)}$		1.5	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector - base cutoff current	3036	Bias condition D; V_{CE} = 30 V dc	I_{CBO3}		100	μA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE3}			
2N1131, L				10		
2N1132, L				15		

See footnote at end of table.

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* TABLE I. Group A inspection.

* Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short circuit forward-current transfer ratio	3206	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc; f = 1 kHz	h_{fe1}			
2N1131, L				15	50	
2N1132, L				30	90	
Small-signal short circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 5.0 mA dc; f = 1 kHz	h_{fe2}			
2N1131, L				20		
2N1132, L				30		
Small-signal open-circuit output admittance	3216	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc	h_{ob1}		1.0	µmho
Small-signal open-circuit output admittance	3216	V_{CE} = 10 V dc; I_C = 5.0 mA dc	h_{ob2}		5.0	µmho
Small-signal short-circuit input impedance	3201	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc	h_{ib1}	25	35	Ω
Small-signal short-circuit input impedance	3201	V_{CE} = 10 V dc; I_C = 5.0 mA dc	h_{ib2}		10	Ω
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CE} = 5.0 V dc; I_C = 1.0 mA dc	h_{rb1}		$8x10^{-4}$	
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CE} = 10 V dc; I_C = 5.0 mA dc	h_{rb2}		$8x10^{-4}$	
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 20 MHz	$\lvert h_{fe} \rvert$			
2N1131, L				2.5	20	
2N1132, L				3.0	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{obo}		45	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1.0 MHz	C_{ibo}		80	pF
Pulse response		(see figure 2)	$t_{on} + t_{off}$		50	ns
Subgroups 5, 6 and 7						
Not applicable						

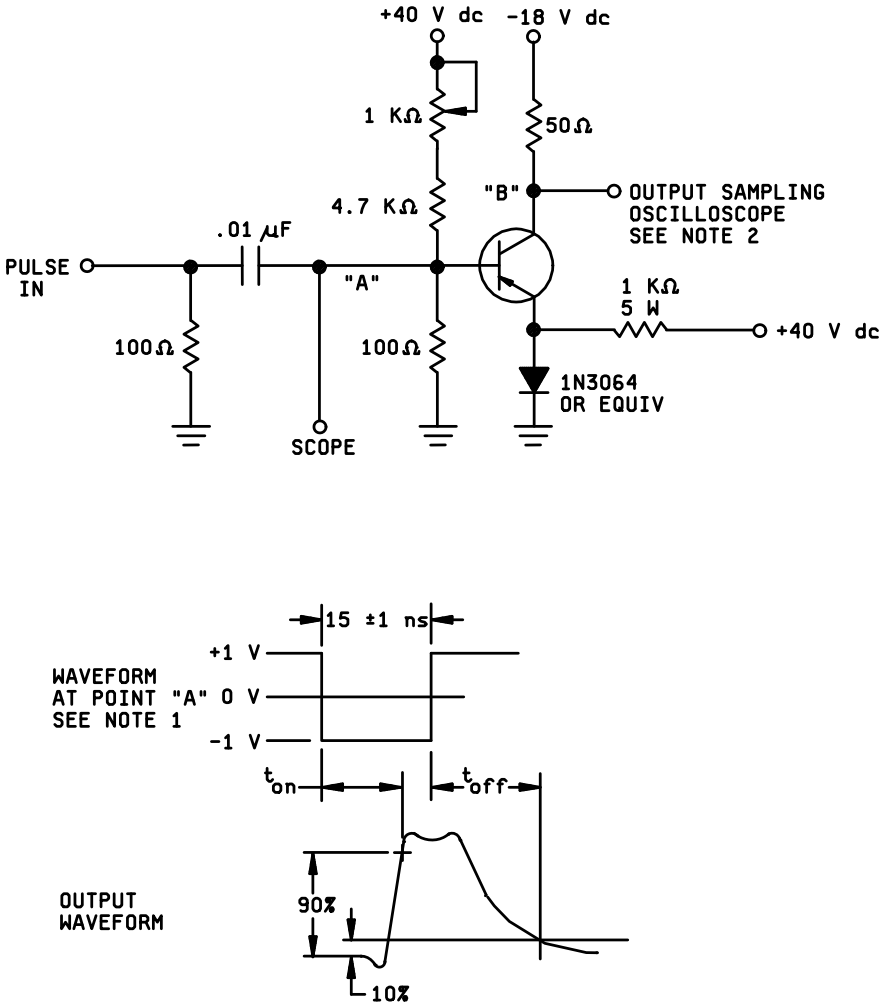
1/ For sampling plan see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			
Thermal impedance curves		See table E-IX of MIL-PRF-19500, group E, subgroup 4.	
Subgroup 5			
Not applicable			
Subgroup 6			
Electrostatic discharge (ESD)	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B for devices < 400 V.	

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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ .5 ns.

FIGURE 2. Nonsaturated switching time test circuit.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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